

EffiDec3D: An Optimized Decoder for High-Performance and Efficient 3D Medical Image Segmentation

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Abstract

Recent 3D deep networks such as SwinUNETR, SwinUNETRv2, and 3D UX-Net have shown promising performance by leveraging self-attention and large-kernel convolutions to capture the volumetric context. However, their substantial computational requirements limit their use in real-time and resource-constrained environments. The high #FLOPs and #Params in these networks stem largely from complex decoder designs with high-resolution layers and excessive channel counts. In this paper, we propose EffiDec3D, an optimized 3D decoder that employs a channel reduction strategy across all decoder stages, which sets the number of channels to the minimum needed for accurate feature representation. Additionally, EffiDec3D removes the high-resolution layers when their contribution to segmentation quality is minimal. Our optimized EffiDec3D decoder achieves a 96.4% reduction in #Params and a 93.0% reduction in #FLOPs compared to the decoder of original 3D UX-Net. Similarly, for SwinUNETR and SwinUNETRv2 (which share an identical decoder), we observe reductions of 94.9% in #Params and 86.2% in #FLOPs. Our extensive experiments on 12 different medical imaging tasks confirm that EffiDec3D not only significantly reduces the computational demands, but also maintains a performance level comparable to original models, thus establishing a new standard for efficient 3D medical image segmentation. Our implementation is available at <https://github.com/SLDGroup/EffiDec3D>.

1. Introduction

Medical image segmentation is a fundamental task in clinical practice, enabling precise delineation of anatomical structures for diagnostic and therapeutic purposes. Deep learning methods have become the gold standard for image segmentation. Among these, the U-shaped CNN architectures [10, 23, 32] have been particularly influential due to

their encoder-decoder structure with skip connections, allowing them to capture both global and local contextual information from 2D images. Building on U-Net’s success, models such as UNet++ [32] and nnU-Net [7] introduce enhancements like redesigned skip pathways and automated configuration, thus achieving high-resolution segmentation across a variety of medical imaging tasks.

Recently, vision transformers [6] have shown promise in segmentation tasks [2, 3, 19–22, 29] by capturing long-range dependencies with self-attention (SA) mechanisms. Hierarchical transformers like Swin Transformer [13] and PVT [28] have advanced multi-resolution feature processing and yielded strong performance in segmentation tasks. However, 2D methods remain limited in capturing the full spatial relationships inherent in 3D volumetric data, as they process 3D data slice-by-slice, resulting in a loss of contextual information along the third dimension.

To overcome these limitations, researchers have developed 3D architectures that process volumetric data more effectively. Early 3D models, such as 3D U-Net [4] and nnUNet3D [7], extended the 2D U-Net structure by incorporating 3D convolutions, enabling the extraction of spatial features along all three axes. Recent advances include transformer-based 3D models such as UNETR [8], nnFormer [31], TransBTS [27], SwinUNETR [26], and SwinUNETRv2 [9] that use self-attention to capture long-range dependencies. Similarly, 3D UX-Net [12] and MedNeXt [24] are introduced to mimic the long-range capabilities of transformers using large-kernel convolutions within a CNN-based architecture. These models make significant progress in 3D medical image segmentation; however, they substantially increase #Params and #FLOPs, which limits their applicability in real-time settings and resource-constrained environments, such as hospitals with limited GPU resources or mobile and embedded systems, where memory and computing power are often limited. UNETR++ [25], SegFormer3D [18], and SlimUNETR [16] introduce efficiency-focused architectures, but at the expense of suboptimal segmentation accuracy.

A common challenge in 3D architectures, especially in models like UNETR [8] and its derivatives, is the high complexity of their decoders. These models typically utilize a large number of channels that increase as the resolution decreases, resulting in a rapid rise in #Params and #FLOPs (e.g., 3D UX-Net [12], SwinUNETR [26], and SwinUNETRv2 [9]). Additionally, these decoders often incorporate high-resolution convolutional operations during the reconstruction stages, which add only marginal accuracy improvements, but significantly increase #FLOPs and #Params (see Figure 1a).

To address the computational challenges inherent in 3D medical segmentation architectures, we propose an *optimized decoder*, EffiDec3D, designed to reduce both #Params and #FLOPs while maintaining segmentation performance. As illustrated in Figure 1b, these optimizations drastically cut down the computational costs, enabling better applicability in real-time or resource-constrained scenarios. For instance, as shown in Figure 2, our EffiDec3D (1.8 MParams, 43.7 GFLOPs) achieves a 96.4% reduction in #Params and a 93.0% reduction in #FLOPs compared to the original decoder of 3D UX-Net (51.7 MParams, 624.2 GFLOPs). In summary, our contributions are as follows:

- **Complexity Analysis of SOTA Architectures:** We perform an in-depth analysis of popular 3D medical segmentation architectures, including 3D UX-Net, SwinUNETR, and SwinUNETRv2, to identify key bottlenecks in #FLOPs and #Params. This analysis highlights the contribution of high-resolution layers and excessive #channels to high computational costs, thus informing our design choices for an optimized decoder.
- **Design of the Optimized EffiDec3D Decoder:** Based on our complexity analysis, we propose *EffiDec3D*, an optimized decoder architecture that achieves a 96.4% reduction in #Params and a 93.0% reduction in #FLOPs for the 3D UX-Net network, and similar reductions of 94.9% in #Params and 86.2% in #FLOPs for SwinUNETR and SwinUNETRv2, without significantly compromising segmentation accuracy (see Figure 2).
- **Adaptable Channel Reduction and Resolution Strategy:** *EffiDec3D* reduces the #channels across decoder stages and selectively removes the high-resolution layers. Consequently, this approach balances computational efficiency with performance and thus can be tuned for different segmentation tasks and architectures.
- **Extensive Multi-Model and Multi-Task Evaluation:** We validate *EffiDec3D* across multiple segmentation architectures (i.e., 3D UX-Net [12], SwinUNETR [26], and SwinUNETRv2 [9]) on 12 diverse medical imaging datasets. Our evaluations demonstrate the robustness and generalizability of our optimized decoder across a variety of segmentation models and clinical datasets.
- **Efficiency without Compromising Segmentation Per-**

formance: We demonstrate that high-FLOP, high-resolution blocks can be minimized or even removed with minimal impact on segmentation accuracy, as evidenced by the high DICE scores. This makes *EffiDec3D* particularly suitable for deployment in resource-constrained environments where computational efficiency is critical.

The remaining of this paper is organized as follows: Section 2 summarizes related work. Section 3 describes the proposed method. Section 4 explains our experimental setup and shows results on 12 medical image segmentation benchmarks. Section 5 covers different ablation experiments. Lastly, Section 6 concludes the paper.

2. Related Work

2.1. 2D Medical Image Segmentation Methods

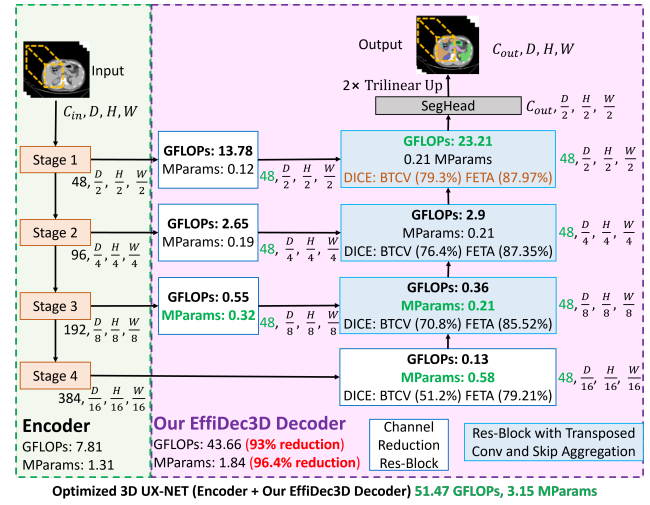
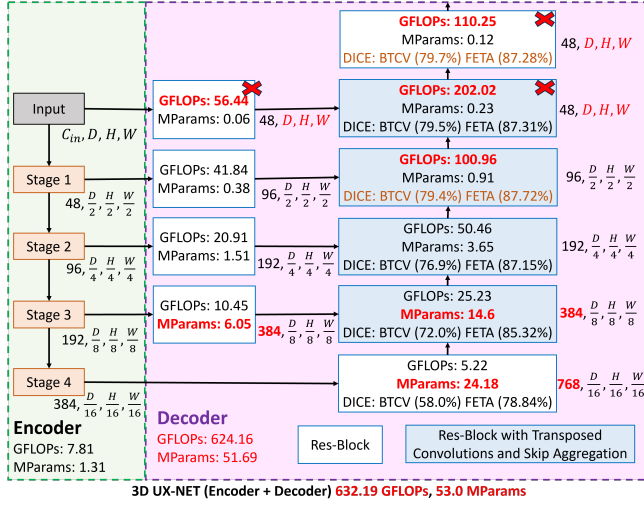
Convolutional Neural Networks (CNNs) have long been the cornerstone of medical image segmentation. The U-Net architecture [23], with its encoder-decoder structure and skip connections, was a pivotal advancement, providing a powerful way to capture both local and global contextual information. Extensions of U-Net, such as UNet++ [32] and ResUNet [30] improve the segmentation accuracy, but at the cost of increased computational complexity. nnU-Net [7], on the other hand, automates the adaptation of U-Net configurations to various datasets, thus advancing U-Net’s generalizability across different datasets. Despite their effectiveness, these models struggle to handle 3D volumetric data, as they process images slice-by-slice, thus potentially missing essential spatial relationships in the third axis.

Vision transformers [6] have recently shown promise in segmentation by capturing global pixel relationships through self-attention. TransUNet [3] integrates CNNs for local feature extraction with transformers for global context, effectively enhancing both local and global feature extraction. Swin-Unet [2] utilizes Swin Transformer blocks [13] in a U-shaped architecture, advancing segmentation performance through hierarchical feature extraction at various scales. Other works, such as PolypPVT [5], CASCADE [19], and EMCAD [22], combine CNN-based decoders with transformer encoders to balance computational efficiency with contextual awareness. However, all of these approaches have limitations in 3D segmentation where the spatial context along all three dimensions is critical.

2.2. 3D Medical Image Segmentation Methods

3D CNN-based models such as 3D U-Net [4] and nnUNet3D [7] adapt the U-shaped structure by incorporating 3D convolutions, thus allowing these networks to capture spatial information along all three axes. This advancement proved valuable for volumetric tasks where contextual information across slices is essential.

Recent advancements in 3D transformer-based models



(a) #FLOPs and #Params analysis of 3D UX-Net [12] with respect to DICE score. Blocks with significantly high #FLOPs and #Params are highlighted in red. This analysis reveals that a high number of channels contribute to high #FLOPs and #Params, while high-resolution convolution blocks (H, W, D) drive up #FLOPs. $\times$ represents the high resolution blocks with very high #FLOPs that could be removed with little to no loss in the DICE score.

(b) Optimized 3D UX-Net with our EffiDec3D decoder. In this design, each decoder block is reduced to 48 channels (i.e., the minimum across all encoder stages), thus resulting in a significant reduction in both #Params and #FLOPs. Additionally, the $\times$ marked (in (a)) high-resolution (H, W, D) blocks are removed, further reducing #FLOPs. The #FLOPs are reported for a 3D volumetric input of resolution $96 \times 96 \times 96$.

Figure 1. Our architecture (a) analysis and (b) optimization. The original decoder in 3D UX-Net has 624.16 GFLOPs and 51.69 Million Parameters (MPParams), whereas our optimized EffiDec3D decoder has only 43.66 GFLOPs (-93.0%) and 1.84 MPParams (-96.4%).

include UNETR [8], nnFormer [31], SwinUNETR [26], and SwinUNETRv2 [9]. These models use self-attention mechanisms to capture long-range dependencies, thus achieving SOTA results on several medical imaging benchmarks. By employing hierarchical transformers like Swin Transformer [13] in the encoder and residual convolutions in the decoder, they capture multi-resolution global and local information effectively. Similarly, 3D UX-Net [12] and MedNeXt [24] introduce large-kernel convolutions in place of self-attention layers to achieve similar performance gains while maintaining a CNN-based structure. However, these 3D architectures significantly increase the FLOP and parameter counts, especially in the decoder, which presents challenges in terms of computational efficiency. More recently, efficiency-focused architectures such as UNETR++ [25], SegFormer3D [18], and SlimUNETR [16] have been developed, at the cost of low segmentation accuracy.

Our work builds upon these findings and focuses on the bottlenecks in the decoder design. In models such as SwinUNETR [26], SwinUNETRv2 [9], and 3D UX-Net [12], the decoder accounts for a substantial part of the computation due to repeated convolutional operations in the high-resolution stages and large #channels across stages. While these designs improve segmentation accuracy, they also limit model applicability to faster training and inference.

Starting from these limitations, our approach leverages a systematic channel reduction strategy and selectively removes high-resolution layers when they provide minimal benefit. This ensures efficient computation while preserving the 3D segmentation performance.

3. Methodology

In this section, we introduce EffiDec3D, which is applicable across various backbone encoders, including both CNN-based (e.g., 3D UX-Net) and transformer-based encoders (e.g., SwinUNETR and SwinUNETRv2). EffiDec3D uses channel reduction and strategic removal of high-resolution convolutions to minimize the #FLOPs and #Params.

3.1. Complexity Analysis of SOTA Architectures

A) **High FLOPs in High-Resolution Decoder Stages:** The decoder stages that operate at full input resolution ($H \times W \times D$) or half-resolution ($\frac{H}{2} \times \frac{W}{2} \times \frac{D}{2}$) are the primary contributors to the high FLOPs in these models. At these resolutions, the need to process large spatial dimensions with substantial channel depths results in a large number of computations. For instance:

- **FLOP Analysis at ($H \times W \times D$):** As the decoder attempts to reconstruct finer details at the highest resolution, the computations scale significantly due to the cubic growth of spatial dimensions in 3D. This leads to extensive FLOP counts, particularly in the last few stages (see Figure 1a), where the upsampling path requires not only dense convolutions, but also transposed convolutions and skip connections. Upsampling followed by residual convolutions from ($\frac{H}{2} \times \frac{W}{2} \times \frac{D}{2}$) to ($H \times W \times D$) alone can require several hundred GFLOPs, which limits the feasibility of these models for real-time applications.
- **Impact on DICE Scores:** In our experiments (see Fig-

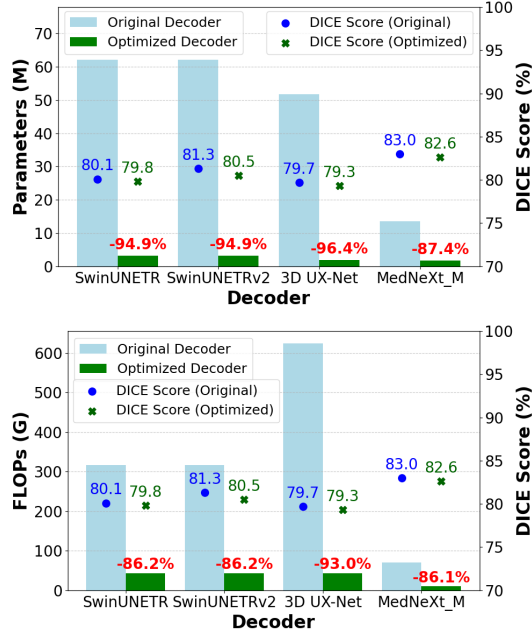


Figure 2. Comparison of decoder computational complexity between original and optimized methods (as described in Section 3.2). #Params (top) and #FLOPs (bottom) for different methods with average DICE score on BTCV multi-organ segmentation dataset. As shown, our optimized decoder significantly reduces #FLOPs and #Params with minimal loss in DICE scores.

ure 1a), we observed that reducing the output resolution to $(\frac{H}{2} \times \frac{W}{2} \times \frac{D}{2})$ instead of $(H \times W \times D)$ yields minimal segmentation accuracy loss, but leads to significant computational savings. Specifically:

- * On BTCV dataset, this reduction decreases the DICE score by a nominal 0.3% (from 79.7% to 79.4%) while reducing 368.71 GFLOPs.
- * On FeTA dataset, this approach results in an accuracy gain of 0.44% (from 87.28% to 87.72%) with similarly high #FLOPs savings.

B) Excessive Channel Counts and Parameter Overhead:

The decoders in 3D UX-Net, SwinUNETR, and SwinUNETRv2 utilize large channel counts, particularly in residual convolution blocks, to capture spatial features at different scales. However, this high parameter count does *not* necessarily translate to proportionate improvements in segmentation accuracy and often results in substantial parameters and computational costs:

- **Residual Convolutions with Large #Channels:** For example, a residual block with 768 channels can contribute approximately 24.8M parameters in a single layer. These large channel sizes in decoder, though could be useful for feature refinement, drastically increase #Params without proportional gains in accuracy.
- **Residual Convolutions and Transposed Convolutions with Large #Channels:** Another significant parameter overhead arises from skip connections that

integrate features across encoder-decoder stages with upsampling. For instance, a residual convolution (12.24M) and a transposed convolution (2.36M) involving 768 input channels and 384 output channels can add 14.6M parameters.

Key Insights and Optimization Directions: To sum up, the computational cost in these existing 3D segmentation models is heavily influenced by both high-resolution decoder stages and excessive channel configurations within residual convolution blocks. Our findings indicate that targeting these factors through careful channel reduction and limiting upsampling resolution can lead to substantial computational savings without significantly impacting segmentation quality. This analysis forms the basis for the design of our optimized EffiDec3D decoder.

3.2. Design of the EffiDec3D Decoder

We propose *EffiDec3D* (as illustrated in Figure 1b), an *optimized decoder* that minimizes the computational overhead while preserving the segmentation accuracy. Below, we discuss each design component of EffiDec3D.

3.2.1 Optimization Strategy and Overall Design

EffiDec3D operates as a *lightweight decoder* by maintaining a reduced channel count across stages and excluding the highest resolution layers. Given a 3D input volume $\mathbf{X} \in \mathbb{R}^{B \times C \times D \times H \times W}$, where B is the batch size, C is the number of input channels, and (D, H, W) represent depth, height, and width, respectively, the encoder processes the input into hierarchical feature maps at varying resolutions. The proposed decoder leverages these feature maps in a low-complexity reconstruction pathway, significantly reducing the #Params and #FLOPs requirements.

Channel Reduction Strategy: To achieve a minimal computational footprint, EffiDec3D employs a uniform channel reduction strategy at each decoding stage using the *ChannelReductionResidualBlock* in Eq. 2. This strategy reduces all feature channels to a constant value, $C_{reduced}$, calculated as the minimum channel count across the encoder stages (Eq. 1):

$$C_{reduced} = \min_{i=1, \dots, n} C_i \quad (1)$$

where C_i represents the number of channels in encoder stage i . By setting $C_{reduced}$ to the minimum viable channel count, EffiDec3D minimizes the computational requirements without significantly compromising the segmentation performance. Each feature map from the encoder, $\mathbf{F}_i \in \mathbb{R}^{B \times C_i \times D_i \times H_i \times W_i}$, is processed as in Eq. 2:

$$\mathbf{F}'_i = \text{ChannelReductionResidualBlock}(\mathbf{F}_i) \quad (2)$$

where $\mathbf{F}'_i \in \mathbb{R}^{B \times C_{reduced} \times D_i \times H_i \times W_i}$. This block ensures that the #channels are reduced to $C_{reduced}$ across all decoding stages, effectively reducing #Params and #FLOPs.

Resolution Restriction and Upsampling: EffiDec3D further optimizes the decoding pathway by limiting the upsampling upto a lower resolution, $\frac{D}{2} \times \frac{H}{2} \times \frac{W}{2}$. Unlike conventional decoders, which upsample to the highest resolution (D, H, W) , our method avoids computationally expensive high-resolution operations that contribute only marginally to performance. The upsampling is conducted using the *ResidualUpBlock* in Eq. 3, which incorporates skip connections from the encoder at corresponding resolutions.

Given the lower-resolution high-level latent feature maps $\mathbf{F}'_i$, the upsampling process integrates skip connections and progressively increases the spatial resolution until reaching $\frac{D}{2} \times \frac{H}{2} \times \frac{W}{2}$. The feature integration at each decoder stage is performed as in Eq. 3:

$$\mathbf{Dec}_i = \text{ResidualUpBlock}(\mathbf{F}'_i, \mathbf{Dec}_{i+1}) \quad (3)$$

where $\mathbf{Dec}_{i+1}$ is the feature map from the previous decoder stage corresponding to $(i + 1)$ encoder stage and $\mathbf{Dec}_n = \mathbf{F}'_n$ (n is the last stage of the encoder).

Final Output Prediction: The final segmentation map is generated from the $(\frac{D}{2}, \frac{H}{2}, \frac{W}{2})$ resolution output using a single prediction head. A $1 \times 1 \times 1$ convolutional layer maps the final decoder output to the desired number of output classes (C_{out}) as in Eq. 4:

$$\mathbf{Y} = \text{Conv3D}(\mathbf{Dec}_i, \mathbf{K}_{1 \times 1 \times 1}) \quad (4)$$

where $\mathbf{K}_{1 \times 1 \times 1}$ is a kernel size of $1 \times 1 \times 1$, and $\mathbf{Y}$ represents the output segmentation map. Finally, we use $2 \times$ trilinear upsampling to get (D, H, W) resolution output (i.e., same resolution as the original input).

3.2.2 ChannelReductionResidualBlock

The *ChannelReductionResidualBlock* (Figure S1 in Supplementary document) is central to our channel reduction approach. By ensuring a constant channel count at $C_{reduced}$ across all decoding stages, this block effectively reduces the model #Params and #FLOPs. Given an input feature map $\mathbf{X} \in \mathbb{R}^{B \times C_i \times D' \times H' \times W'}$, this block applies two successive convolutions (Eqs. 5 and 6):

$$\mathbf{Y}_1 = \text{ReLU}(\text{IN}(\text{Conv3D}(\mathbf{X}, \mathbf{K}, C_{reduced}))) \quad (5)$$

$$\mathbf{Y}_2 = \text{ReLU}(\text{IN}(\text{Conv3D}(\mathbf{Y}_1, \mathbf{K}, C_{reduced}))) \quad (6)$$

where $\mathbf{K}$ is a convolution kernel (i.e., $3 \times 3 \times 3$). IN and $C_{reduced}$ are the Instance Norm and number of output channels, respectively. A residual connection adjusts the input using Eq. 7 if $C_{in} \neq C_{reduced}$, otherwise $X_{residual} = X$:

$$\mathbf{X}_{residual} = \text{IN}(\text{Conv3D}(\mathbf{X}, \mathbf{K}_{1 \times 1 \times 1}, C_{reduced})) \quad (7)$$

$$\mathbf{Y}_{reduced} = \text{ReLU}(\mathbf{Y}_2 + \mathbf{X}_{residual}) \quad (8)$$

where $\mathbf{Y}_{reduced}$ is the output feature map of this block.

3.2.3 ResidualUpBlock

In *ResidualUpBlock*, the lower-resolution decoder feature $\mathbf{Dec}_{i+1}$ is first upsampled using a transposed convolution $\mathbf{U}_i = \text{TransposedConv}(\mathbf{Dec}_{i+1})$, where $\mathbf{U}_i \in$

$\mathbb{R}^{B \times C_{reduced} \times D_i \times H_i \times W_i}$ is the upsampled feature map with the same spatial resolution as the corresponding skip feature map $\mathbf{F}'_i$. The upsampled feature map $\mathbf{U}_i$ is either concatenated or added to the skip feature map $\mathbf{F}'_i$ as in Eq. 9, with addition (+) offering reduced complexity.

$$\mathbf{Dec}_i = \mathbf{U}_i + \mathbf{F}'_i \quad \text{or} \quad \mathbf{Dec}_i = \text{Concat}(\mathbf{U}_i, \mathbf{F}'_i) \quad (9)$$

To be noted, in case of concatenation, the number of output channels will be the sum of the #channels from both $\mathbf{U}_i$ and $\mathbf{F}'_i$. The resulting feature map $\mathbf{Dec}_i$ is refined using a residual convolutional block as in Eqs. 5, 6, 7, and 8.

4. Experiments

In this section, we provide a detailed evaluation of our optimized 3D decoder across multiple datasets. **The dataset descriptions, implementation details, computational metrics, additional results, and more ablation studies are provided in the Supplementary document.**

4.1. FeTA 2021 Segmentation Results

In Table 1, our optimized architectures, incorporating the EffiDec3D decoder, demonstrate substantial improvements in computational efficiency while maintaining or even improving the segmentation accuracy across various brain structures. Specifically, integrating EffiDec3D with 3D U-Net reduces the parameter count by over 94% (from 53M to 3.16M) and GFLOPs by 92% (from 631.97M to 51.47M) with a DICE score increase from 87.28% to 87.97%. This highlights our optimized decoder's effectiveness in improving efficiency without compromising performance. Similarly, integrating EffiDec3D with SwinUNETR and SwinUNETRv2 results in over 83% reductions in both #Params and #FLOPs, with DICE scores remaining competitive. For SwinUNETRv2, EffiDec3D achieves an increase in average DICE (from 87.29% to 87.91%).

A detailed comparison reveals that although performance on structures like External Cerebrospinal Fluid (ECF) and Deep Grey Matter (DGM) shows minor declines, substantial gains are observed in larger, well-defined structures, especially the Cerebellum (Cereb.), where the DICE scores increase by over 4% across all three optimized models. This improvement suggests that EffiDec3D's channel reduction and selective upsampling effectively capture features for large, structured regions, while the minor trade-offs in more complex, smaller regions like DGM and ECF likely stem from the reduced model capacity.

Overall, Table 1 highlights the efficiency of our optimized models across a range of brain structures, with strong performance on larger, clear regions, making them useful for computationally efficient medical segmentation tasks.

4.2. BTCV 13-organ Segmentation Results

Table 2 shows the validation DICE scores for 13 organ segmentation tasks in the BTCV dataset. Our optimized

Architecture	MParams ↓	GFLOPs ↓	ECF	GM	WM	Vent.	Cereb.	DGM	BS	Avg. (%)↑	HD95 ↓
SlimUNETR [16]	1.79	<u>20.17</u>	87.44	79.29	92.1	87.76	79.46	86.59	86.87	85.64	2.83
SegFormer3D [18]	4.50	5.03	88.16	79.63	92.59	89.17	83.50	85.82	87.10	86.57	2.70
UNETR [8]	92.78	82.60	86.56	72.98	91.72	85.64	86.60	86.86	78.99	84.19	3.41
nnUNet [7]	31.78	417.96	85.97	78.56	90.82	81.87	88.64	81.16	84.96	84.57	3.12
nnFormer [31]	149.25	213.60	88.48	81.12	93.06	90.41	81.36	87.21	87.55	87.03	2.72
TransBTS [27]	31.58	110.34	88.57	80.48	92.78	88.74	84.91	86.81	87.99	87.18	2.48
UNETR++ [25]	42.62	53.99	88.61	81.42	93.03	88.87	80.6	86.78	87.71	86.72	2.87
MedNeXt.M.K3 [24]	17.55	110.65	88.87	81.18	93.06	90.47	85.95	87.15	87.53	87.74	2.43
MedNeXt.M.K3 w/ EffiDec3D (Ours)	5.77	49.73	88.46	81.45	93.03	90.05	83.46	88.10	88.23	87.54	2.48
3D UX-Net [12]	53.00	631.97	88.61	81.26	<u>93.09</u>	90.42	82.26	87.23	88.11	87.28	2.54
3D UX-Net w/ EffiDec3D (Ours)	<u>3.16</u>	51.47	88.32	81.26	93.10	<u>90.52</u>	<u>86.55</u>	87.39	88.68	87.97	<u>2.38</u>
SwinUNETR [26]	69.19	337.61	88.65	80.83	92.88	90.06	80.44	86.34	87.79	86.71	2.33
SwinUNETR w/ EffiDec3D (Ours)	11.21	57.29	88.34	80.95	92.93	90.16	85.35	87.24	87.55	87.50	2.45
SwinUNETRv2 [9]	83.19	353.61	<u>88.86</u>	81.22	93.02	90.29	81.63	<u>87.73</u>	88.25	87.29	2.44
SwinUNETRv2 w/ EffiDec3D (Ours)	18.21	63.29	<u>88.77</u>	<u>81.31</u>	93.04	90.56	86.03	<u>87.37</u>	<u>88.30</u>	<u>87.91</u>	2.41

Table 1. Fetal brain segmentation DICE scores (%) and average HD95 distances on the FeTA 2021 test dataset. The #FLOPs are reported for a one-channel volume of $96 \times 96 \times 96$. The original MedNeXt.M.K3, 3D UX-Net, SwinUNETR, and SwinUNETRv2 are compared side by side with our EffiDec3D-optimized versions. **Bold** and underlined are the best and second-best values. **Note:** ECF: External Cerebrospinal Fluid, GM: Grey Matter, WM: White Matter, Vent.: Ventricles, Cereb.: Cerebellum, DGM: Deep Grey Matter, BS: Brainstem.

Architecture	Spl	RKid	LKid	Gall	Eso	Liv	Sto	Aor	IVC	Veins	Pan	Rad	Lad	Avg. (%)↑	HD95↓
SlimUNETR [16]	84.88	81.79	83.05	64.63	65.39	94.61	76.13	87.34	80.08	58.96	57.37	49.34	46.45	71.54	12.34
SegFormer3D [18]	87.42	82.72	84.92	70.82	69.42	93.83	79.05	87.82	81.30	63.18	64.47	52.76	48.75	74.34	13.41
UNETR [8]	88.36	82.86	84.10	66.28	70.88	95.07	78.27	86.82	79.93	63.41	63.68	58.43	56.41	74.96	21.78
nnUNet [7]	88.00	84.78	85.47	71.85	75.44	95.32	76.54	90.93	85.14	68.91	62.19	66.9	60.19	77.82	10.91
nnFormer [31]	92.40	83.31	85.39	72.02	73.16	94.85	83.61	89.56	81.64	67.63	70.76	61.58	61.71	78.28	10.57
TransBTS [27]	89.92	84.31	85.59	73.86	72.09	96.20	81.49	89.81	85.23	65.67	71.20	63.74	66.15	78.87	14.47
UNETR++ [25]	94.69	85.99	86.90	78.83	73.79	96.22	83.27	91.2	87.02	72.40	71.82	67.64	62.19	80.92	9.59
MedNeXt.M.K3 [24]	90.08	86.96	88.96	77.27	78.16	96.91	84.24	92.16	88.65	75.45	80.13	68.83	70.87	82.98	5.45
MedNeXt.M.K3 w/ EffiDec3D (Ours)	90.81	<u>86.49</u>	86.60	79.90	<u>76.68</u>	<u>96.87</u>	<u>84.17</u>	<u>91.75</u>	<u>88.40</u>	<u>73.05</u>	<u>79.35</u>	<u>70.42</u>	68.61	<u>82.55</u>	<u>6.41</u>
3D UX-Net [12]	92.47	84.39	86.54	78.72	74.16	95.44	82.47	90.93	85.03	70.56	64.6	66.49	64.85	79.74	12.43
3D UX-Net w/ EffiDec3D (Ours)	90.44	84.79	<u>87.12</u>	75.26	74.31	94.35	78.79	90.98	85.89	68.04	70.15	65.52	64.67	79.25	10.12
SwinUNETR [26]	88.56	85.92	86.03	79.40	75.50	95.41	79.58	90.13	86.18	71.12	69.36	69.35	65.19	80.13	14.01
SwinUNETR w/ EffiDec3D (Ours)	<u>92.93</u>	83.50	85.73	76.83	74.03	95.31	79.15	91.04	85.62	69.33	71.38	67.24	65.62	79.82	8.47
SwinUNETRv2 [9]	90.78	86.29	85.62	79.20	75.90	95.21	78.90	90.00	86.25	72.61	74.50	71.44	69.66	81.26	12.86
SwinUNETRv2 w/ EffiDec3D (Ours)	89.91	85.79	85.88	78.53	75.71	95.74	77.20	91.55	86.23	71.66	74.35	67.14	67.08	80.52	10.13

Table 2. BTCV 13-organ segmentation validation DICE scores (%) and HD95 distances. The original MedNeXt.M.K3, 3D UX-Net, SwinUNETR, and SwinUNETRv2 are compared side by side with our optimized versions. **Bold** and underlined are the best and second-best values. **Note:** Spl: spleen, RKid: right kidney, LKid: left kidney, Gall: gallbladder, Eso: esophagus, Liv: liver, Sto: stomach, Aor: aorta, IVC: inferior vena cava, Veins: portal and splenic veins, Pan: pancreas, Lad: left adrenal glands, Rad: right adrenal glands.

EffiDec3D-based models, when incorporated into MedNeXt, 3D UX-Net, SwinUNETR, and SwinUNETRv2, achieve substantial reductions in #Params and #FLOPs while achieving comparable or improved DICE scores across most organs. For example, our optimized 3D UX-Net achieves a competitive average DICE score (79.25%) compared to the original (79.74%), with a huge reduction in computational complexity (94% fewer #Params and #FLOPs). Similarly, our optimized SwinUNETR maintains an average DICE score of 79.82%, close to the original’s 80.13%, while being more efficient. SwinUNETRv2 w/ EffiDec3D achieves an average DICE score of 80.52%, slightly lower than the original SwinUNETRv2 (81.26%), indicating minimal trade-offs in performance.

Examining individual organ DICE scores reveals that certain organs, such as spleen, kidneys, and liver, generally maintain high accuracy across both original and optimized models, likely due to their relatively larger size and clearer anatomical boundaries. However, there are slight

drops in performance for small or thin structures like esophagus (Eso) and veins, where precise segmentation is more challenging and may be affected by the lower output resolutions in our optimized models. Interestingly, 3D UX-Net with EffiDec3D sees improvements in complex regions like IVC and pancreas (Pan), suggesting that our optimizations are effective at retaining critical features for more anatomically challenging structures.

4.3. MSD (10 Datasets) Segmentation Results

In Table 3, our EffiDec3D-optimized versions of 3D UX-Net, SwinUNETR, and SwinUNETRv2 demonstrate strong average performance across the 10 MSD datasets, achieving competitive or improved DICE scores despite significant reductions in computational complexity (upto 94% reduction in #Params and #FLOPs). These optimizations are particularly beneficial for resource-constrained and real-time applications, as they retain much of the segmentation accuracy while drastically lowering the computational costs.

Architecture \ Dataset	Task01_BrainTumour				Task02	Task03_Liver			Task04_Hippocampus			Task05_Prostate		
	TC	WT	NET	Avg.	Heart	Liver	Cancer	Avg.	Ant.	Post.	Avg.	PZ	TZ	Avg.
SlimUNETR [16]	79.86	87.95	50.18	72.66	90.42	89.77	44.47	67.12	87.19	85.38	86.29	28.07	71.22	49.65
SegFormer3D [18]	79.14	88.15	54.26	73.85	91.64	87.56	42.55	65.06	87.44	85.48	86.46	51.06	80.43	65.75
UNETR [8]	79.77	89.83	57.47	75.69	91.42	81.49	39.75	60.62	88.01	86.34	87.18	35.70	76.96	56.33
nnFormer [31]	80.09	88.38	55.89	74.79	92.21	92.24	58.14	75.19	87.58	85.84	86.71	45.87	80.90	63.39
TransBTS [27]	83.19	90.14	60.15	77.83	90.12	91.09	60.73	75.91	88.39	86.68	87.54	43.83	77.06	60.45
UNETR++ [25]	81.6	90.26	59.62	77.16	92.39	92.53	57.40	74.97	88.51	87.01	87.76	32.56	74.91	53.74
3D UX-Net [12]	82.9	91.13	61.72	78.58	92.03	92.36	65.82	79.09	89.33	87.64	88.49	48.99	82.67	65.83
3D UX-Net w/ EffiDec3D (Ours)	82.34	90.88	60.96	78.06	92.63	92.22	64.58	78.40	88.89	87.20	88.05	56.97	82.87	69.92
SwinUNETR [26]	83.19	91.36	62.62	79.06	91.92	92.92	63.45	78.19	88.61	87.12	87.87	49.21	76.64	62.93
SwinUNETR w/ EffiDec3D (Ours)	83.36	91.15	61.86	78.79	91.98	93.55	62.58	78.07	88.56	87.06	87.81	48.89	80.32	64.61
SwinUNETRv2 [9]	83.44	91.08	61.82	78.78	91.96	91.89	61.28	76.59	88.48	86.86	87.67	52.03	81.31	66.67
SwinUNETRv2 w/ EffiDec3D (Ours)	83.78	90.85	61.02	78.55	92.29	93.68	63.16	78.42	88.71	86.77	87.74	51.95	82.84	67.40

Architecture \ Dataset	Task06_Lung	Task07_Pancreas			Task08_HepaticVessel			Task09	Task10_Colon	10 Tasks
	Cancer	Pan.	Cancer	Avg.	Vessel	Tumour	Avg.	Spleen	Cancer	Avg. (%)
SlimUNETR [16]	67.66	59.88	25.64	42.76	48.88	33.52	41.20	95.42	25.13	63.83
SegFormer3D [18]	55.07	44.83	18.15	31.49	37.11	29.24	33.18	95.68	33.09	63.13
UNETR [8]	65.38	52.94	20.35	36.65	55.67	36.94	46.31	94.60	28.03	64.22
nnFormer [31]	69.79	68.44	33.28	50.86	55.79	53.33	54.56	91.40	24.74	68.36
TransBTS [27]	63.57	69.43	34.34	51.89	56.40	51.00	53.70	94.54	29.53	68.51
UNETR++ [25]	76.09	68.98	35.03	52.01	56.81	46.44	51.63	96.49	40.82	70.31
3D UX-Net [12]	71.46	68.75	33.81	51.28	58.09	53.20	55.65	96.68	48.77	72.79
3D UX-Net w/ EffiDec3D (Ours)	74.14	70.06	34.40	52.23	56.40	54.84	55.62	96.56	48.42	73.40
SwinUNETR [26]	65.12	65.60	32.30	48.95	58.62	45.84	52.23	96.13	42.94	70.53
SwinUNETR w/ EffiDec3D (Ours)	73.79	68.57	36.39	52.48	58.63	48.21	53.42	96.55	45.41	72.29
SwinUNETRv2 [9]	73.52	76.90	38.27	57.59	60.39	59.27	59.83	96.77	48.87	73.83
SwinUNETRv2 w/ EffiDec3D (Ours)	75.19	75.36	42.82	59.09	59.90	58.83	59.37	96.86	52.23	74.71

Table 3. Results of MSD 10 datasets. We report the DICE scores (%) excluding the background class on the validation dataset. Original 3D UX-Net, SwinUNETR, and SwinUNETRv2 are compared side by side with our EffiDec3D-optimized versions. Our ‘SwinUNETRv2 with EffiDec3D’ achieves the 10 tasks average best DICE score (74.71%). Best values are in **Bold**. Note: TC: Tumour Core, WT: Whole Tumour, NET: Non-enhancing Tumour, Pan.: Pancreas, Ant.: Anterior, Post.: Posterior, PZ: Peripheral Zone, TZ: Transition Zone.

In several cases, EffiDec3D provides consistent improvements in accuracy. For example, in Task05_Prostate, 3D UX-Net with EffiDec3D achieves an average DICE score of 69.92%, higher than the original 3D UX-Net’s 65.83%. Such improvements indicate that our optimized decoder effectively captures essential spatial information without relying on excessive channel or high-resolution operations, making it well-suited for complex anatomical segmentation.

Our optimized models also show efficient performance gains without major accuracy losses in key tasks. For instance, SwinUNETRv2 with EffiDec3D outperforms the original SwinUNETRv2 in Task03_Liver, moving from 76.59% to 78.42% in average DICE. Similarly, SwinUNETR with EffiDec3D improves the original SwinUNETR in Task09_Spleen, increasing from 96.13% to 96.55%. These results reflect that EffiDec3D’s optimization, particularly by minimizing channel usage and restricting high-resolution layers, is capable of capturing sufficient contextual information needed for accurate segmentation.

However, there are minor losses in specific tasks involving finer structures, where slightly reduced spatial resolution impacts performance. For example, in Task10_Colon, 3D UX-Net with EffiDec3D shows a small drop in DICE

score from 48.77% to 48.42%. This reduction could be attributed to the optimized decoder’s inability to fully capture fine details in smaller or more complex anatomical structures due to reduced channels or upsampling.

5. Ablation Studies

We conduct a series of ablation studies¹ to evaluate the effect of various design choices in our optimized 3D decoder.

5.1. Effect of Removing High-Resolution Stages

Table 4 presents a performance and computational complexity comparison across different output resolutions for a channel-optimized 3D UX-Net model on the BTCV dataset. As expected, maintaining full resolution (D, H, W) achieves the highest average DICE score (79.44%). However, this comes at a substantial cost: 404.4 GFLOPs and 3.55 MParams. Reducing the output resolution to (D/2, H/2, W/2) significantly lowers the computational load to 51.47 GFLOPs with a minimal drop in segmentation performance (79.25% average DICE score). This balance between accuracy and computational efficiency makes (D/2, H/2, W/2) a compelling choice for applications with limited resources.

¹More ablation studies are provided in the Supplementary document.

Output Resolution	MParams ↓	GFLOPs ↓	Spl	RKid	LKid	Gall	Eso	Liv	Sto	Aor	IVC	Veins	Pan	Rad	Lad	Avg. (%)↑
Original (D, H, W)	53	632.19	92.47	84.39	86.54	78.72	74.16	95.44	82.47	90.93	85.03	70.56	64.6	66.49	64.85	79.74
D, H, W	3.55	404.4	91.31	85.47	85.68	72.2	76.01	95.06	80.04	90.99	86.02	72.44	69.24	61.9	66.31	79.44
D/2, H/2, W/2 (Ours)	3.15	51.47	90.44	84.79	87.12	75.26	74.31	94.35	78.79	90.98	85.89	68.04	70.15	65.52	64.67	79.25
D/4, H/4, W/4	2.82	14.41	86.83	83.27	84.29	73.43	72.66	95.14	78.54	90.81	84.34	66.43	66.78	55.71	55.1	76.41
D/8, H/8, W/8	2.41	8.85	83.9	81.34	83.43	65.54	67.44	93.56	72.09	87.01	81.34	56.64	61.67	45.84	40.18	70.77
D/16, H/16, W/16	1.88	7.94	72.68	68.41	64.99	40.93	44.3	90.65	57.19	61.38	59.68	26.17	37.37	30.8	11.33	51.22

Table 4. Performance comparison at different output resolutions with corresponding computational complexity (MParams, GFLOPs) and DICE scores (%) for each organ on BTCV dataset. ‘Original (D, H, W)’ is the non-optimized 3D UX-Net for reference. For other rows, we use our channel-optimized (for a fair comparison) 3D UX-Net network for this set of experiments; hence the small number of parameters.

Decoder #Channels	MParams ↓	GFLOPs ↓	DICE (%)↑
16	1.73	14.83	77.25
24	2.01	21.19	78.69
32	2.34	29.47	78.47
40	2.72	39.65	78.73
48 (Ours)	3.15	51.47	79.25
56	3.52	53.73	79.46
64	3.93	56.23	78.67
72	4.84	61.94	78.35
80	4.37	58.97	78.77
88	5.35	65.14	79.41
96	5.88	68.45	79.21
Original	53	632.19	79.74

Table 5. Impact of decoder channel count on performance for the BTCV dataset with our optimized 3D UX-Net architecture. The #Params and #FLOPs are reported for the full architecture.

As the resolution continues to decrease to (D/8, H/8, W/8), and further to (D/16, H/16, W/16), the DICE scores drop more dramatically, with average scores of 70.77% and 51.22%, respectively. For instance, critical organs like spleen, right kidney, and liver maintain relatively high DICE scores even at (D/8, H/8, W/8) resolution (above 81%), but smaller or more challenging organs, like pancreas, veins, and adrenal glands, see significant performance degradation. This trend underscores that lower resolutions may inadequately capture fine details, especially for smaller structures, which limits the segmentation accuracy.

In summary, reducing the output resolution is an effective strategy for balancing segmentation performance and computational demands. (D/2, H/2, W/2) resolution offers an optimal middle ground, preserving high accuracy for larger organs while maintaining a low computational footprint. Lower resolutions, however, lead to diminished performance, particularly in finer anatomical structures, thus highlighting the importance of resolution for capturing essential details in medical imaging tasks.

5.2. Effect of Varying the Decoder #Channels

We examine the impact of varying the number of channels in the decoder by comparing the segmentation performance for different reduced #channels in our decoder. Table 5 shows the effect of varying the decoder #channels on the segmentation performance and computational complexity of our optimized 3D UX-Net network on the BTCV dataset. We can see a key trade-off between model complexity (MParams and GFLOPs) and DICE score.

As the number of decoder channels increases from 16 to 96, the model’s computational load (both MParams and GFLOPs) generally rises, reaching 5.88M parameters and 68.45 GFLOPs for 96 channels. However, this increased complexity does not always lead to substantial performance gains in terms of DICE score. For instance, with 48 channels, the model achieves a DICE score of 79.25%, which is close to the best DICE score (79.74% with the original, unoptimized model) in Table 5, but with significantly fewer #Params and #FLOPs. This stability of the DICE score suggests diminishing returns in performance when increasing the #channels beyond a certain point. Our optimized decoder with 48 or 56 channels appears to achieve an effective balance between computational efficiency and segmentation accuracy, achieving nearly the original model’s performance (79.74%) with far fewer #Params and #FLOPs.

Interestingly, the 88-channel model achieves one of the highest DICE scores at 79.41%, but this marginal improvement over 48 or 56 channels comes with significantly increased #Params and #FLOPs. This finding suggests that using 48 or 56 channels could be optimal, as they deliver competitive segmentation performance with a fraction of the computational cost of the original 3D UX-Net decoder.

6. Conclusion

In this paper, we have introduced EffiDec3D, a highly optimized decoder that addresses the computational inefficiencies in 3D medical image segmentation by reducing channel counts and eliminating high-resolution layers in the decoder. Our approach achieves substantial reductions in #FLOPs and #Params (up to 93.0% and 96.4%, respectively, in 3D UX-Net) with minimal to no decrease in segmentation accuracy, as demonstrated on 12 segmentation tasks.

Our optimized decoder is suitable for integration with various backbone networks for 3D segmentation, and our optimization strategy offers a generalizable solution for refining computationally demanding other decoders. This work highlights the potential of decoder optimization in advancing efficient, scalable medical imaging models.

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